An Exploratory Phenomenological Study of a Local Variable-Scale Propagator for Galactic Rotation Curves

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ABSTRACT

We present an **exploratory phenomenological study** of a non-local velocity modification model based on a variable-scale propagator. By postulating a relaxation scale $l(r) = l_0 / V_{bar}^2(r)$ that varies inversely with the local Newtonian baryonic potential, we test the effective velocity relation $V_{eff}^2(r) = V_{bar}^2(r) + B_0^2 \cdot exp(-l(r)/r) / (l(r)/r)$ across disk galaxies. Evaluated on the **SPARC dataset (175 disk galaxies)** using two fixed scale parameters ($l_0 = 36{,}034.48\ km^2 s^{-2} kpc$ and $B_0^2 = 1.00\ km^2 s^{-2}$), the model achieves an unweighted global RMSE of **38.78 km/s**. However, rigorous analytical examination reveals a fundamental limitation: in the asymptotic regime ($r \to \infty$), the model predicts a logarithmic slope $d\ ln\ V_{flat} / d\ ln\ M = (1 + \eta)/2$ (where $\eta = l_0 / (GM)$), which approaches 0.50 when $\eta \ll 1$. This diverges from the empirical slope of ≈ 0.25 required by the Baryonic Tully-Fisher Relation (BTFR). We present this formulation as an exploratory benchmark and detail the mathematical origin of this asymptotic boundary.

1. INTRODUCTION & MOTIVATION

The flattening of galactic rotation curves **Lelli et al. (2016)** and the tightness of the Baryonic Tully-Fisher Relation (BTFR) **Lelli et al. (2019)** continue to challenge our understanding of dark matter halos versus modified gravity frameworks **Milgrom (1983)**. Phenomenological models provide a testable middle ground by exploring how the spatial distribution of visible baryons might directly regulate galactic dynamics.

2. PHENOMENOLOGICAL PROPAGATOR FORMULATION

The model posits a non-local quadratic modification to the Newtonian rotation velocity $V_{bar}(r)$:

$$V_{eff}^2(r) = V_{bar}^2(r) + B_0^2 \cdot \exp(-l(r)/r) / (l(r)/r)$$

where $l(r)$ is a local relaxation scale modulated inversely by the baryonic potential $V_{bar}^2(r)$:

$$l(r) = l_0 / V_{bar}^2(r) \quad [\text{kpc}]$$

The parameters and physical dimensions are defined as:

- $l_0 = 36{,}034.48\ \text{km}^2\text{s}^{-2}\text{kpc}$: Universal scale constant.
- $B_0^2 = 1.00\ \text{km}^2\text{s}^{-2}$: Propagator amplitude squared.

3. EMPIRICAL ASSESSMENT ON THE SPARC DATABASE

The formulation was evaluated against all 175 disk galaxies in the SPARC rotation curve database without individual galaxian parameter tuning.

175 / 175	38.78 km/s	3,345	1.289
GALAXIES ANALYZED	GLOBAL RMSE	DATA POINTS	M87* SCALE H



Figure 1: Schematic behavior of the variable-scale propagator as a function of the local baryonic potential depth.

4. DIMENSIONLESS SCALE AT COMPACT REGIMES

For a point-mass distribution representing a supermassive black hole such as M87* **Event Horizon Telescope (2019)** with mass $M \approx 6.5 \times 10^9 M_\odot$ (yielding $GM \approx 27{,}956\ km^2 s^{-2} kpc$), the dimensionless scale parameter evaluates to:

$$\eta = {}^{l_0}/_{GM} = {}^{36{,}034.48}/_{27{,}956} \approx 1.289$$

This confirms that l_0 maintains a consistent order of magnitude across galactic and compact scales.

Dataset / Target	Sample Size (N)	Data Points	Metric	Value
Full SPARC Catalogue	175	3,345	Global RMSE	38.78 km/s
Low Baryonic Potential Subgroup	89	1,258	Subgroup RMSE	30.71 km/s
Compact Object (M87*)	1	Horizon Scale	Dimensionless Ratio η	1.289

5. LIMITATIONS AND ASYMPTOTIC BEHAVIOR

For a finite baryonic mass distribution of total mass M, the Newtonian velocity at large distances satisfies $V_{bar}^2(r) \simeq GM/r$. Substituting this into the model yields:

$$\eta = {l(r)}/{r} = {l_0}/{GM} \quad (\text{constant with respect to } r)$$

The asymptotic flat velocity V_{flat} then takes the exact form:

$$V_{flat}^2 = B_0^2 \, {e^{-\eta}}/{\eta} = B_0^2 \, {GM}/{l_0} \, e^{-l_0/(GM)}$$

Differentiating $\ln V_{flat}$ with respect to $\ln M$ yields the predicted logarithmic slope:

$$d \ln V_{flat} / d \ln M = {1 + \eta}/{2}$$

In the limit $\eta \ll 1$ (for massive galaxies where $GM \gg l_0$), this slope approaches $1/2$ (corresponding to $M_{bar} \propto V_{flat}^2$). In contrast, the empirically observed Baryonic Tully-Fisher Relation (BTFR) requires $M_{bar} \propto V_{flat}^4$ **Lelli et al. (2019)**, which implies a slope of $\approx 1/4$ (or 0.25) when expressing velocity as a function of mass.

This divergence represents a critical mathematical boundary of the current formulation. While the model captures local velocity profiles over intermediate radial ranges (1–30 kpc), its unmodified asymptotic tail fails the BTFR power-law test at spatial infinity.

6. CONCLUSION

This work documents a testable exploratory model based on a variable-scale propagator. While achieving an unweighted fit of 38.78 km/s across SPARC, analytical evaluation reveals that the current formulation cannot account for the BTFR slope at large radii. Future research must determine whether a generalized relaxation function can reconcile this asymptotic slope while preserving verifiable empirical performance.

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